Roll No: _____

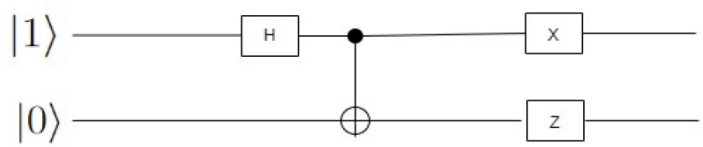
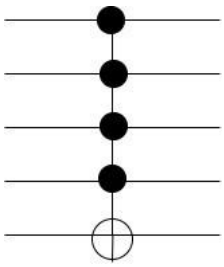
Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
END Semester Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours (2:00 pm-5:00 pm)
Marks : 40

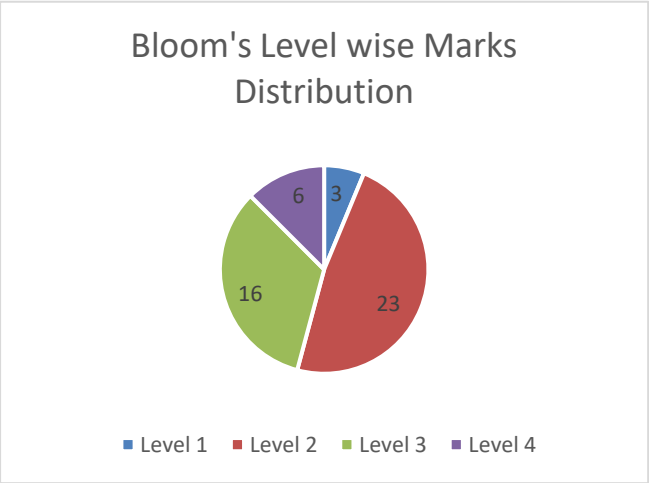
Course Code: UCS619
Date: May 13, 2024
Name of Faculty: Dr. Ajay Kumar, Dr Samya Muhuri

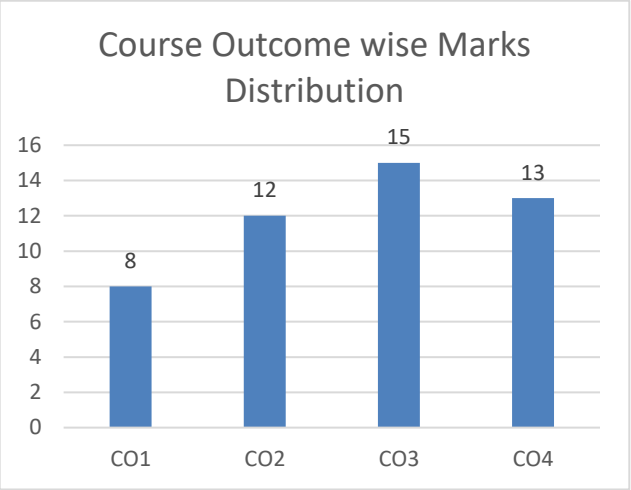
Note: Attempt any **5** questions. Assume missing data, if any, suitably. Give appropriate reasoning for your answers.

Q.No	Questions	Marks	CO	BL
1.a.	Given the search space | 000 >, | 001 >, ..., | 111 >. Design the Grover's algorithm circuit (with the detailed oracle and Grover diffusion operator) to find the element 110 in the search space. Explain the value at each point for searching the element 110.	5	CO4	L2
1.b.	Using the appropriate assumptions, describe any three postulates of Quantum Mechanics in detail.	3	CO1	L1
2.a.	Consider the following scenario in BB84 protocol that is used in quantum cryptography. Suppose Alice prepares a qubit on a Computational basis. Prove the following using the measurement concept for the following cases: **Case 1:** If Bob measures the qubit on a computational basis. Prove that the measurement probabilities of the qubit at Bob's end are 1. Case 2: If Bob measures the qubit on a Hadamard basis. Prove that the measurement probabilities of the qubit at Bob's end are $\frac{1}{2}$.	5	CO3	L3
2.b.	Design the circuit for the entangled quantum systems $\frac{1}{\sqrt{2}}(\|011\rangle - \|100\rangle)$.	3	CO2	L3
3.a.	Deutsch-Jozsa algorithm is used to check whether a function is constant or bal- anced. Draw the circuit of Deutsch's algorithm and explain the value at each point for the following scenario of three qubits: $f(000) = f(010) = f(100) = f(110) = 1, f(001) = f(011) = f(101) = f(111) = 0$.	6	CO4	L2
3.b.	Explain the advantages of the Deutsch-Jozsa algorithm over the classical approach for checking whether a function is constant or balanced.	2	CO4	L2
4.a.	Design the circuit for correcting phase-flips and bit-flips using the Shor's 9-qubit code.	3	CO2	L3

4.b.	Consider a Y error occurs at 2nd qubit. Explain the status of quantum system of each step of the Shor's 9-qubit code.	5	CO3	L2
5.a.	The matrix product $$|a\rangle \otimes |b\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} -2 \\ 1 \\ 2 \\ -1 \end{pmatrix}$$ and $$|a\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$ Find $|b\rangle$.	3	CO1	L3
5.b.	What is the output at each step of the following circuit? 	3	CO2	L4
5.c.	Show that the Fredkin gate can act as NOT Gate.	2	CO1	L3
6.a.	Write down various steps of Shor's algorithm. Explain the procedure for factoring N=15. Please note that you need to explain the concept of Periodic function.	5	CO3	L2
6.b.	Using CCNOT or Toffoli gate implement the following circuit. 	3	CO2	L4

Marks Distribution


Bloom's Level wise Marks Distribution
3
23
16
6
Level 1
Level 2
Level 3
Level 4


Course Outcome wise Marks Distribution
16
14
12
10
8
6
4
2
0
8
12
15
13
CO1
CO2
CO3
CO4